

DATA TYPES

1)Create a console application to accept the side of a square and print the area of the square.

```
using System;

class Program
{
    static void Main(string[] args)
    {
        Console.Write("Enter the side of the square: ");
        double side = double.Parse(Console.ReadLine());
        double area = Math.Pow(side, 2);
        Console.WriteLine("Area of the square = " + area);
    }
}
```

OUTPUT

Enter the side of the square: 6

Area of the square = 36

2)Create a console application that finds the number of digits and alphabets in a given string.

```
using System;

class Program
{
    static void Main(string[] args)
```

```
{  
    Console.Write("Enter a string: ");  
    string str = Console.ReadLine();  
    int alphabets = 0, digits = 0;  
    for (int i = 0; i < str.Length; i++)  
    {  
        char ch = str[i];  
        if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z'))  
        {  
            alphabets++;  
        }  
        else if (ch >= '0' && ch <= '9')  
        {  
            digits++;  
        }  
    }  
    Console.WriteLine("Number of Alphabets = " + alphabets);  
    Console.WriteLine("Number of Digits = " + digits);  
}  
}
```

OUTPUT

Enter a string: Nandini20

Number of Alphabets = 7

Number of Digits = 2

3)Create a console application to increment each character by one and toggle its case.

using System;

class Program

{

static string ConvertString(string str)

{

string result = "";

foreach (char ch in str)

{

if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z'))

{

char c = (char)(ch + 1);

if (c >= 'A' && c <= 'Z')

result += (char)(c + 32);

else if (c >= 'a' && c <= 'z')

result += (char)(c - 32);

}

else

{

result += ch;

```

        }
    }
    return result;
}

static void Main(string[] args)
{
    Console.Write("Enter a string: ");
    string str = Console.ReadLine();
    Console.WriteLine("Output: " + ConvertString(str));
}
}

```

OUTPUT

Enter a string: ab12cd34

Output: BC12DE34

4) Create a console application to accept a string and perform the below string manipulation on the same:

- 1. Print the string in reverse order.**
- 2. Extract part of the string from 2nd position till the end of the string.**
- 3. Replace any given character by '\$' and print the new string.**
- 4. Copy the string to another string variable, modify the data in the 2nd string variable and print both the strings.**

using System;

```
class Program
{
    static void Main()
    {
        Console.Write("Enter a string: ");
        string str = Console.ReadLine();

        // 1. Reverse the string
        string reverse = "";
        for (int i = str.Length - 1; i >= 0; i--)
        {
            reverse = reverse + str[i];
        }

        Console.WriteLine("Reverse String: " + reverse);

        // 2. Extract from 2nd position till end
        string sub = str.Substring(1);
        Console.WriteLine("Substring: " + sub);

        // 3. Replace character with '$'
        Console.Write("Enter character to replace: ");
        char ch = Convert.ToChar(Console.ReadLine());
        string replaced = str.Replace(ch, '$');

        Console.WriteLine("After Replace: " + replaced);

        // 4. Copy string and modify second string
        string str2 = str;
```

```
        str2 = str2 + " welcome";  
        Console.WriteLine("Original String: " + str);  
        Console.WriteLine("Modified Copied String: " + str2);  
    }  
}
```

OUTPUT

Enter a string: Nandini

Reverse String: inidnaN

Substring: andini

Enter character to replace: i

After Replace: Nand\$n\$

Original String: Nandini

Modified Copied String: Nandini welcome